

Triangle Congruence: SSS, SAS, ASA, AAS and HL

Answer Key

High-School Geometry

Quick Answers

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|------|-------|-------|-------|
| 1. — | 3. 15 | 5. 12 | 7. — |
| 2. 4 | 4. 67 | 6. — | 8. 12 |
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Curriculum

Level: High-School Geometry

HSG-CO.B – *problems 1, 2, 3, 4, 5, 6, 8*

HSG-CO.C – *problem 7*

Difficulty 1–5 of 5 across 13 tagged checks.

1. Name the evidence. Read each list by *position*, not just by count — the same three letters can be a criterion or nothing at all. (Short written answers: open response, compare with the table below.)

	Criterion	Why
(a)	SSS	all three pairs of sides
(b)	ASA	the side $\overline{AB}$ lies <i>between</i> $\angle A$ and $\angle B$
(c)	SAS	$\angle B$ lies <i>between</i> $\overline{AB}$ and $\overline{BC}$
(d)	AAS	$\overline{BC}$ is <i>not</i> between $\angle A$ and $\angle B$
(e)	HL	right angles, congruent hypotenuses, one congruent leg
(f)	none	$\angle A$ is not between $\overline{AB}$ and $\overline{BC}$: this is SSA

Row (f) is the one to dwell on. It has two sides and an angle, exactly like row (c), but the angle sits outside the pair of sides, and that single difference is the whole reason (c) proves congruence and (f) does not (see problem 6).

2. A shared side. The ticks give $\overline{AB} \cong \overline{CB}$, so their lengths are equal: $5x - 3 = 2x + 9$. Subtract $2x$ from both sides: $3x - 3 = 9$; add 3: $3x = 12$; divide: $\boxed{x = 4}$. (Check: $5(4) - 3 = 17$ and $2(4) + 9 = 17$.)

Criterion (open response). $\triangle ABD \cong \triangle CBD$ by **SSS**. The three pairs are $\overline{AB} \cong \overline{CB}$ (single ticks), $\overline{AD} \cong \overline{CD}$ (double ticks), and $\overline{BD} \cong \overline{BD}$. That third pair is not marked because it is the *same segment* in both triangles; the reflexive property of congruence supplies it. A shared side or shared angle is almost always the missing third pair in a two-triangle figure.

3. Vertical angles. $\angle AEB$ and $\angle CED$ are vertical angles, so they are congruent and their measures are equal: $4x + 10 = 6x - 20$. Subtract $4x$: $10 = 2x - 20$; add 20: $30 = 2x$; divide: $x = 15$. (Check: $4(15) + 10 = 70$ and $6(15) - 20 = 70$.)

Criterion (open response). $\triangle ABE \cong \triangle CDE$ by **SAS**. The pairs are $\overline{AE} \cong \overline{CE}$ (single ticks), $\angle AEB \cong \angle CED$, and $\overline{BE} \cong \overline{DE}$ (double ticks). The vertical-angle theorem supplies the angle pair, and that angle sits *between* the two marked sides, which is what makes this SAS rather than SSA.

4. ASA or AAS? The angles of a triangle sum to a straight angle, so $m\angle C = 180^\circ - 52^\circ - 61^\circ$. Solving $x + 52 + 61 = 180$ gives $x = 180 - 113$, so $m\angle C = 67^\circ$.

Which criterion (open response). As marked, the congruent side $\overline{BC}$ is opposite $\angle A$, so it is *not* between the two marked angles: that arrangement is **AAS**.

Once the third angle is known, $\angle B$ and $\angle C$ are both angles of $\triangle ABC$ and $\overline{BC}$ lies between them, so the same figure also gives **ASA**. Both work because two angles determine the third: whenever a triangle has AAS evidence it automatically has ASA evidence as well, which is why AAS is a theorem rather than a separate postulate.

5. Hypotenuse–Leg. $\triangle ABC$ has its right angle at C , so $\overline{AB}$ is the hypotenuse and the Pythagorean theorem gives $BC = \sqrt{AB^2 - AC^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144}$, so $BC = 12$.

Why HL works (open response). The calculation used only the hypotenuse and one leg — once those two are fixed, the third side is forced; there is no choice left. So in $\triangle DEF$, which has the same hypotenuse and the same marked leg, EF must also be 12. The two triangles therefore have all three pairs of sides congruent, and SSS finishes the argument. HL is a shortcut through SSS that only right triangles get, which is why it is the one legal arrangement that looks like SSA.

6. Why SSA is not a criterion (open response — compare to this). In the figure, $\triangle ARP$ and $\triangle ARQ$ share $\angle A$, share the side $\overline{AR}$, and have $\overline{RP} \cong \overline{RQ}$. That is two sides and a non-included angle in both triangles — SSA evidence. But $\triangle ARQ$ is visibly bigger: $\overline{AQ}$ is longer than $\overline{AP}$, and $\angle APR$ is obtuse while $\angle AQR$ is acute. Two non-congruent triangles satisfy the same SSA data, so SSA cannot be a criterion.

Geometrically, the swinging side of fixed length reaches the far ray in *two* places, so the data does not pin the triangle down. **What rescues it:** require the angle to be a right angle and the swinging side to be the hypotenuse — then the arc meets the ray only once, and the arrangement becomes HL. (Knowing the triangle is obtuse at the third vertex, or that the side opposite the given angle is the longer one, also removes the ambiguity.) Accept any answer that names the two possible triangles and gives one such extra condition.

7. Two-column proof (open response — model answer).

Statements	Reasons
1. $\overline{AB} \parallel \overline{DC}$, $\overline{AD} \parallel \overline{BC}$	1. Given ($ABCD$ is a parallelogram)
2. $\angle BAC \cong \angle DCA$	2. Alternate interior angles, $\overline{AB} \parallel \overline{DC}$, transversal $\overline{AC}$
3. $\angle BCA \cong \angle DAC$	3. Alternate interior angles, $\overline{AD} \parallel \overline{BC}$, transversal $\overline{AC}$
4. $\overline{AC} \cong \overline{CA}$	4. Reflexive property of congruence
5. $\triangle ABC \cong \triangle CDA$	5. ASA (steps 2, 4, 3)

The diagonal is doing two jobs: it is the transversal that produces both pairs of alternate interior angles, and it is the shared side that supplies the third pair. Note the side $\overline{AC}$ lies between the two angle pairs used, so the criterion is ASA. A student who lists the parts in the order $\angle$, $\angle$, side without checking position may claim AAS; here the side is included, so ASA is the accurate name. Accept any correct proof, including one that first proves opposite sides congruent and finishes with SSS.

8. Judge the claim. $\angle R \cong \angle U$ means their measures are equal: $3x + 8 = 5x - 16$. Subtract $3x$: $8 = 2x - 16$; add 16: $24 = 2x$; divide: $\boxed{x = 12}$. (Check: $3(12) + 8 = 44$ and $5(12) - 16 = 44$.)

Judging Rico (open response). He is wrong. The given sides are $\overline{PQ}$ and $\overline{QR}$, which meet at Q , so the angle *between* them is $\angle Q$ — but the given angle is $\angle R$, which sits at the far end of $\overline{QR}$. The arrangement is therefore **SSA**, not SAS, and problem 6 shows SSA does not force congruence. The letters are the same; the position is not.

One extra given that fixes it. Any of these works: $\angle Q \cong \angle T$ (making it genuine SAS); $\overline{PR} \cong \overline{SU}$ (making it SSS); or right angles at P and S with $\overline{PQ}$, $\overline{ST}$ as the hypotenuses (making it HL). Accept any single condition that either supplies the included angle, supplies the third side, or turns the figure into a pair of right triangles.